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Live Estimating the Carbon Footprint of Additively Manufactured Components – a Case Study

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Abstract

The carbon footprint is an important indicator of the effects of climate change and thus a new control variable for the design and production of new products. A prerequisite for managing and controlling the carbon footprint is calculating and visualizing it transparently. In this work, a general framework for live estimating the carbon footprint for production processes is presented. The framework is used to investigate the carbon footprint of additively manufactured components. The novelty of the proposed approach lies in the straightforwardness of the general framework with a holistic data model. With this it is possible to compare different Life Cycle Impact Assessment (LCIA) methods and assumptions of new additively manufactured parts, as all data and methods are integrated from data acquisition at the machine to balancing compactly. A demonstrator is developed that calculates the consumption of energy and materials in real-time. The demonstrator consists of a AM machine, sensors for measuring all consumption, data processing in the back end, and a user-friendly dashboard in the front end. Due to different assumptions and calculation methods, there is a high variation in the final result of the carbon footprint calculation, which is presented in this paper.

Keywords: Carbon Footprint; Live Estimation; LCIA; Additively Manufactured Components; Sustainability

1. Introduction

For some years, the industry has been increasingly aware of the need to produce in a more climate-friendly manner. Companies are therefore progressively evaluating their Corporate Carbon Footprint (CCF) and their Product Carbon Footprint (PCF) based on suitable Life Cycle Impact Assessment (LCIA) methodologies to reduce these through various measures. With the growing introduction of smart machines equipped with information and communication technology for Industry 4.0, more and more real-time data can be accessed and processed to ensure transparent accounting of the required resources. However, for most companies it is easier to publish the global, annual consumption and emissions than to provide the product-specific inputs and outputs or even a live balancing with real-time access, as it is proposed by scientific works [1].

Nevertheless, in the context of Industry 4.0, all machine data is already being collected, processed, and visualized in real-time. This is essential in the age of digitalization since products can be customized to an increasing level and thus rigid production according to certain patterns is no longer appropriate. Furthermore, continuous machine data monitoring enables early detection of malfunctions due to abrasion and, as a result, it is possible to react in due time to avoid expensive downtimes.

By combining the two aspects of real-time data and LCIA methods, the decision quality concerning the environmental sustainability of machines and products can be increased. This work presents a systematic framework for estimating the carbon footprint of process machines. The overall approach is presented in a case study of additively manufactured components by designing a new demonstrator for live estimation of the carbon footprint. Primary data from various sensors are recorded to obtain trustworthy inputs for LCIA calculation which is directly integrated into the back end. Besides systematically adding data acquisition and processing to the additive manufacturing (AM) machine, a dashboard that allows changing assumptions and different LCIA calculation methods is developed. The result of the changes can be seen in a live graph for the current carbon footprint of the produced product.

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2. State of the art

For the live data acquisition of production machines, the ETA factory of the Technical University of Darmstadt can be referred to as a pioneering example. To determine and ultimately minimize the energy demand of the process chain, more than 2,000 data points are already being continuously recorded in real time by sensors, evaluated in parallel, and visualized [2]. Especially the evaluation of energy flows is a major challenge, as they are highly dynamic but not directly visible and mostly distorted by a multitude of time-variable influencing factors. Therefore, all energy flows are systematically recorded, and, through effective energy monitoring, it is possible to give early warning of excessive consumption, for example, caused by malfunctioning equipment or improper use by users.

From the automotive sector, Real Driving Emissions (RDE) can be mentioned as an additional example of a live data application [3]. In addition to conventional certification on the test bench with stationary measurement technology, the combustion engine emissions are determined in real-time with a real driver on the road. A Portable Emission Measurement System (PEMS) is used for this purpose, which determines all data during real driving and evaluates and visually displays it in real-time. With this technology, the measurement of CO_2, CO, NOx, and the particle can be used to fulfill emission regularities.

A further method of monitoring consumption data is by using Digital Twins. [4] presents how Digital Twins can be used to create energy-efficient manufacturing by a holistic review, the presentation of Digital Twins frameworks, and methodologies for energy consumption at the machine process level.

Monitoring of complex systems can also be done by the combination of sustainability indicators like the carbon footprint with the Internet of Things (IoT) technology. [5] use real-time data to monitor an entire port in a single tool by calculating specific environmental Key Performance Indicators (KPIs). In the absence of standards for comparing different complex systems such as ports, the transparent approach is intended to enable comparison of the "environmental performance" of ports. Another example of monitoring complex systems can be found in [6]. There, statistical process monitoring is presented as an effective tool for monitoring and controlling carbon emissions from industrial facilities. With this tool, it is possible to save costs and carbon emissions as well as perform early maintenance or repair.

[7] presents a framework for live LCA for 3D-printed products by combining Industry 4.0 methods and global warming potential (GWP) calculations. The result is a dashboard where a production manager can track GWP values while printing a new product. Design and printing parameters can be changed to minimize the GWP.

When using LCA for the carbon footprint calculation, a main problem is the comparability of studies using different methods and assumptions [8]. Even in established standards differences can be found [9]. Examples of the variation of the carbon footprint can be seen in the production of a particleboard [10] (-692 to 433 $kg(CO_2eq)/m^3$) or in the production of table potatos [11] (0.10 to 0.16 $kg(CO_2eq)/kg$)

Global research on the carbon footprint as a new control variable in a wide range of sectors shows its importance as an ecological indicator. Data acquisition is usually not the problem, but rather the conversion to the stand-alone value of the carbon footprint.

Compared to the work of [7] where a live LCA for 3D printed products was developed, this paper gives a more general approach for live estimating the carbon footprint for all production machines, focusing on the estimation of the final value. The data is acquired directly in the process and not from estimating in the printing software like it is done for the needed material in [7]. In addition, the approach taken in this work is to show how different the final value of the carbon footprint can be and how important it is to be transparent about the communication of its origins like system boundaries, conditions, and calculation methods.

In the context of this work, a straightforward approach to the live estimation of the carbon footprint is developed, which makes it possible to change the conditions and calculations in a transparent and user-friendly way, to sensitize for the variations in the value of the carbon footprint.

3. Concept for the Product Carbon Footprint Calculation

The research in this paper is guided by the Design Research Methodology (DRM) type three [12]. First, the research question is clarified based on a literature review, followed by a research-based descriptive study. The prescriptive study is comprehensively conducted by the general framework in this chapter and the derived application in section 4 and evaluated through an initial descriptive study in section 5.

This chapter provides the general framework for live estimation of carbon footprint in new product manufacturing. Three sequential steps are necessary as shown in Figure 1, which are explained in the following. They are guided by a holistic carbon footprint (CO_2eq) data model. The data model describes all the information needed to calculate the carbon footprint of selected system boundaries and goes beyond the classic approach, such as that used in the Greenhouse Gas Protocol (GHG).

Fig. 1: Steps for the product carbon footprint calculation

3.1. Data Acquisition

The first step to quantify the carbon footprint of new components is to identify all consumers who are necessary to produce the component. Depending on the process and the component, material and energy flows have to be analyzed. Material flows can be divided into material that is part of the product such as raw material for a CNC process or plastics such as granulate or filament for printing or injection molding and materials that are consumed indirectly such as coolants and lubricants. To track the material flows, different sensors have to be integrated into the process such as flow sensors. Besides the material flow, energy consumption has to be measured. The measurement concept varies depending on the type of current (alternating current, three-phase current, and direct current) and the voltage range. The focus of interest when calculating the carbon footprint is the consumption of the entire machine, so one measurement system for the main power line is sufficient. However, it may be of special interest to measure individual modules or actuators to optimize the process. Concerning live processing of the data, all sensors and measuring systems need to have a generally usable interface and an active communication feature.

3.2. Data Transfer

As a second step, the acquired data have to be transferred to the processing unit. The data requires to be available in a digital format. Therefore, for some sensors, an analog-to-digital converter is necessary. The data in digital format can then be sent via a server-client or publish-subscribe network protocol. Examples of server-client structures are Hypertext Transfer Protocol (HTTP) or Open Platform Communications Unified Architecture (OPC UA) and an example of the publish-subscribe structure is the Message Queuing Telemetry Transport (MQTT) protocol. HTTP and MQTT are the most used and adopted protocols [13].

3.3. Data Processing

Data Processing for carbon footprint calculation can be divided into processing primary and secondary data. Primary data are received directly from the sensors which measure the material and energy flows. These are weighted with a specific CO_2 equivalents conversion factor and then summed up for the final result of the current carbon footprint. The conversion factors can be found in the literature or eco-databases such as ecoinvent or GaBi. Secondary data is necessary if the primary data sources are not available or cannot be acquired, e.g. due to information gaps in supply chains. Different calculation methods can be applied to estimate the carbon footprint of the processes which can be used in stand-alone applications such as OpenLCA or even directly integrated into the data processing tool. Direct integration works, for example, via the python framework brightway, in that a complete LCA including LCI databases can be operated via python commands.

4. Application

In the following, the application of the general framework presented in section 3 to an additive manufacturing (AM) process is illustrated. Therefore, a material extrusion (MEX) process is chosen.

4.1. Data Acquisition

There are various energy consumers in the AM machine, which all require a specific amount of power. These include the printing sheet, extrusion nozzle, motors, power supply, control, and display. Besides the main power, the measurement of the printing sheet and the extrusion nozzle is of special interest. The aim is to find out the proportion of consumers that generate heat compared to those that only perform small movements. The measurement is enabled by power sensors for a low voltage range. Similarly, the total current consumption is measured before the power supply is connected. The main power supply is measured using a current clamp and a voltage transformer. The voltage transformer takes the voltage via the socket strip to which the AM machine is connected. The current clamp determines the magnetic flux density via the induction of a wire of the supply cable for the power supply unit. The current is determined by the magnetic flux and the voltage. For the connection of the current clamp, one wire of the power supply cable behind the main switch of the AM machine was laid to the outside. The current clamp can here directly be applied around the wire. The sensor concept is shown schematically in Figure 2.

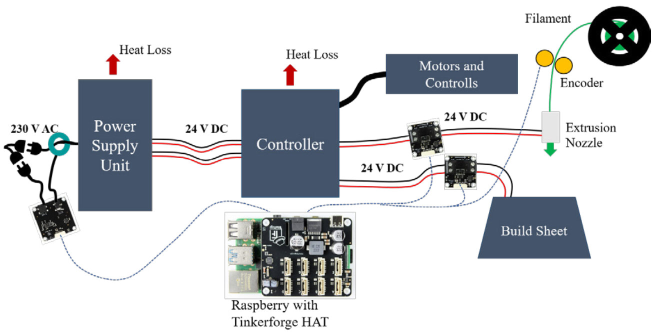

Fig. 2: Sensor Concept for sensing the power and filament consumption of a MEX process

The filament quantity is recorded with the aid of a rotary encoder. A feed roller picks up the movement of the filament through a friction wheel directly located before the extrusion head. The mass of the filament can be determined from the steps of the encoder using a conversion formula. Calibration of the filament measurement is necessary before measurement operation. Due to the friction contact on the driving rollers, the roller engagement radius cannot be measured exactly but must be determined empirically.

4.2. Data Transfer

The sensors described above transmit the measurement data to a Raspberry Pi. A network connection is established between a tablet and the Raspberry Pi using a basic router which allows a client-server structure to be implemented. The OPC UA protocol, which is chosen to be the most suitable for this kind of machine-to-machine communication, is used to transfer the data. The data can thus directly be sent from the Raspberry Pi to the server by a message. This server itself also runs on the

Raspberry Pi. The tablet is the client, reads the data from the server, and processes it further.

4.3. Data Processing

The tablet is also used to process, record, and display the data. In the backend, the database ecoinvent as well as the most common Life Cycle Impact Assessment (LCIA) calculation methods are integrated. The database is used to estimate the conversion factor of the used polylactide (PLA) material. The raw material process and production process for filament (extrusion process) are taken into account for this work. Other materials can also be obtained directly from the database as well as adding own processes. The database and LCIA methods are fully integrated via Python, so there is no need to manually extract the conversion factors from the databases. The conversion factor for energy consumption is taken from literature [14]. 19 different factors from the electricity mix in Germany between 2003 and 2021 are considered.

The CO_2 equivalents from material and energy are summed up to give the latest total consumption. The concept can switch between the parameters on a dashboard to visualize the influences of the different factors and methods (Figure 3).

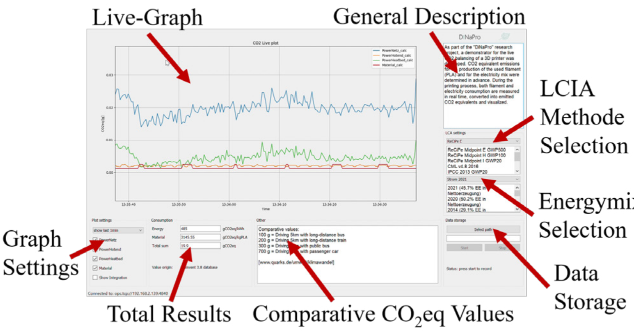

Fig. 3: Dashboard for Live Emissions

The display is then done in a live graph which can either show the converted sensor values directly or integrated. The central component and thus also the largest and most important area for visualization is the live graph, which displays the current live consumption of the Am machine. In the background of the GUI, the recent currents and the encoder position are received via the OPC UA protocol and converted into CO_2 equivalents, which are then plotted over time in the diagram.

A display of the entire time horizon is also possible, which provides the opportunity to recognize both short-term dynamic developments and trends over the recording period. Below this, checkboxes can be used to hide and show individual graphs in the diagram. This allows a detailed view of the development over time of individually selected consumptions. In addition, the current conversion factors are displayed, which can also be changed directly. It is possible to switch between several LCIA calculation methods for the printing material. When the selection is changed, the conversion factors are updated and the carbon footprint of the new consumption is calculated, changing the display both in the live view and in the integration view. Under the currently selected conversion factors, the total consumption since the software connected to the OPC UA server is displayed. The total sum matches the integration view when all individual curves are added up. A picture of the whole setup is displayed in Figure 4.

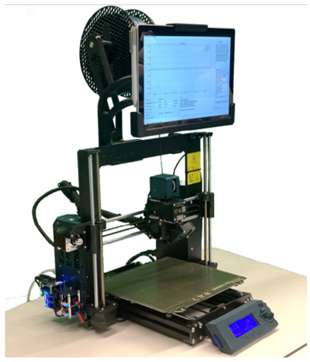

Fig. 4: Demonstrator for Live Emissions

5. Results and discussion

During the recording of the power consumption, a special phenomenon can be observed which contributes significantly to the total consumption. The diagram in Figure 5 shows the raw data of the recorded power consumption of the build sheet, the extrusion nozzle, and the summed-up value of the AM machine from the power supply averaged over one minute. There is a large peak in the graph at the beginning of the recording. Here, the AM machine heats the build sheet to 60 °C and the extrusion nozzle to 215 °C. Afterward, the power drops sharply again which is due to a height calibration run of the AM machine. After this run, the actual printing is carried out and the consumption increases again.

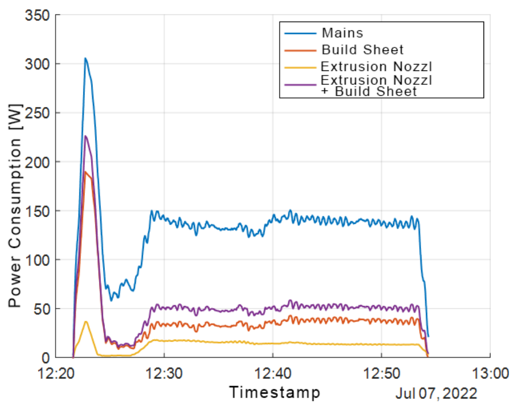

Fig. 5: Raw Data of Energy Consumption

The origin of this peak can be identified as the build sheet, which is part of the total consumption. The cause of the peak is most likely the control parameters of the build sheet, which ensure that the build sheet remains heated to the set 60 °C. It was also noticeable that the percentage contribution of the build sheet and extrusion nozzle has a median value of 30% to the total power consumption. The other 70% of the total consumption is therefore not directly converted into heat. Other possible consumers are the five stepper motors of the axes and the filament feed, as well as the two fans installed at the extrusion nozzle, which immediately cool down the extruded material to make it form-stable. Additional consumers are the built-in display and the AM machine's microcontroller. The losses caused by the AM machine's power supply itself should not be neglected when considering this, because the current clamp which measures the total consumption is still located in front of the power supply. Consequently, the extrusion nozzle and the build sheet likely have a significantly larger effect on the actual consumption of the power supply output but this is not visible in the raw data since the efficiency of the power supply is not taken into account.

Table 1: Extract of the CO_2eq for different energy mixes in germany

electricity mix	gCO_2eq
2021 (45.7% EE)	32.50
2020 (50.2% EE)	29.35
2014 (29.1% EE)	42.81
2010 (18.9% EE)	42.61
2003 (8.6% EE)	47.64

Table 1 shows the carbon footprint of the example component - a gearwheel - for different ratios of renewable energies in the German electricity mix from five different years over the last 20 years. Here, only the energy consumption of the AM machine for the printing process is shown. In the best case, the test component has a CO_2 equivalent emission for the energy consumption of 29.35 g and in the worst case with a low share of renewable energies in the electricity mix an emission of 47.64 g CO_2 equivalent. For the calculation of the total amount of CO_2 equivalent emission, the material consumption has to be added. Therefore, different LCIA methods can be used. In this example, the material for the AM process is PLA (polylactide) - a commonly used material for prototypes. In most cases, the PLA is purchased and the carbon footprint of the material is unknown. The integrated eco-database ecoinvent is used to estimate the carbon footprint of PLA. The carbon footprint of the raw PLA material as well as the production process for filament (extrusion process) is taken into account. The construction of the overall process is based on data from PLA production at NatureWorks' Nebraska facility. However, since the company there owns renewable energy certificates and therefore reduces CO_2 equivalent emissions from the electricity mix used, the process takes into account a representative European UCTE (Union for the Coordination of Transmission of Electricity) electricity mix. The electricity mix is assumed to be fixed in the remainder of the paper because in practice it is often difficult to change the value of the electricity mix of a supplier product.

Table 2 shows the carbon footprint of the example component for the different calculation methods. This includes on one side the material consumption predicted by the slicer software and on the other side the material consumption measured by the rotary encoder which is slightly higher than the prediction from the software. The absolute error between the values varies slightly depending on which calculation method was chosen. The relative error is the same for all methods and amounts to 5.41%. The value from the slicing software is lower, since the test strip, which takes place at the beginning of each new printing process, is probably not taken into account here.

Table 2: CO_2eq with different LCA methods

LCIA-Method	gCO_2eq (slicer)	gCO_2eq (sensor)
ReCiPe Midpoint E-GWP500	4.01	4.22
ReCiPe Midpoint H-GWP100	4.38	4.61
ReCiPe Midpoint I-GWP20	4.89	5.15
CML v4.8 GWP100	4.37	4.61
IPCC 2013 GWP20	5.04	5.31
IPCC 2013 GWP100	4.41	4.65

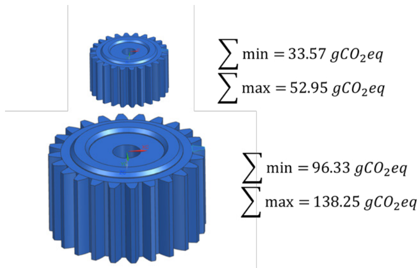

Fig. 6: Minimal and maximal values of the carbon footprint of two types of gearwheels

The total sum of the carbon footprint of the example component has a variance between 33.57 and 52.95 g CO_2 equivalent as it can be seen in Figure 6. It should be noted that for this study five different electricity mixes and six LCIA methods are observed. This range could increase even more if other cases would be integrated such as the futuristic assumption of 100% renewable energy or other LCIA methods. Also, the range increases the larger the component becomes. Scaling to twice the size of the gear results in a minimum carbon footprint value of 96.33 g CO_2 equivalent and a maximum value of 138.25 g CO_2 equivalent. The size of the diameter is not proportional to the variance of the result. In the first case, the variance is 19.38 g CO_2 and in the second case 41.92 g CO_2 which is 216% of the first value. The example component clearly shows that electricity consumption is essentially responsible for the carbon footprint of a component and the influence of the material consumption is smaller than the influence of the energy consumption the larger the component becomes.

6. Conclusion and outlook

Analyzing the raw data from the sensors shows that the heating process at the start of the printing process has a major influence on the overall energy consumption. This influence is most evident for components with a short printing time. Energy consumption and thus the carbon footprint could be minimized if several components were printed at the same time so that the heating process only had to be carried out once.

It is also apparent that the influence of the individual consumers varies - depending on the geometry of the component. The larger the component and thus the printing time, the greater the influence of energy consumption compared to material consumption. It can be stated here that the printing time should be minimized in order to also keep the carbon footprint as small as possible. One possible strategy is to reduce the infill or the printing quality in order to save time. Switching to 100% renewable energy for the AM machine would mean that the large impact of the energy consumer of the printing process could be set to zero. Thus, only the material consumption would be relevant for determining the carbon footprint. This has the consequence that less consideration is given to the reduction of energy consumption. However, reducing energy consumption should be as much a priority as reducing the carbon footprint in the current time. Switching to 100% renewable energy for the production of the PLA would reduce the carbon footprint even more. Here it is often the case that suppliers have to be consulted and it is therefore difficult to influence the value.

The goal of this study was to develop a framework that allows to calculate the carbon footprint of process machines in a transparent and user friendly way. A new tool with a fully integrated LCA calculation and LCI database allows to change parameters of the calculation such as the used energy mix or the LCIA calculation method. It is possible to see the influences when changing parameters on a live graph and can be used without great knowledge in LCIA methods. In addition, all consumptions of the individual components of an AM machine are displayed in detail and can therefore be specifically analyzed. The carbon footprint can consequently be used as a control variable in additive manufacturing.

To expand the concept, a closer examination of the purchased material is required. Different materials as well as the extension of the production process of the materials could be integrated. Likewise, the influence of using different energy generation methods in the upstream processes should be investigated.

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